NEXO MEDICO AI

Enterprise Multi-Tenant Hospital Management & Artificial Intelligence Supercomputer Platform

MAJOR PROJECT TECHNICAL PRESENTATION & REVIEW REPORT

Designed for Team Understanding, Academic Evaluation & Viva Defense

PROJECT AT A GLANCE:

- Core Domain: Healthcare Information Technology & Clinical Artificial Intelligence
- Multi-Tenant Portals: 7 Dedicated Roles (Super Admin, Hospital Admin, Doctor, Nurse, Pharmacist, Lab Tech, Patient)
- AI Engines: 7 Clinical Patient Engines + 6 Super Admin Security & Infrastructure AI Engines
- Multilingual Speech: Real-Time Speech-to-Text & Text-to-Speech in 9+ Languages (English, Telugu, Hindi, Tamil, etc.)
- Tech Stack: Next.js 14 App Router, TypeScript, Tailwind CSS, Prisma ORM, SQLite/PostgreSQL, Web Speech API

Generated for Academic Review | Version 1.0 Production

1. Executive Abstract

Nexo Medico AI is a comprehensive enterprise hospital management platform that integrates traditional hospital administrative workflows (inpatient admissions, bed grid tracking, prescription writing, pharmacy inventory, lab report scanning) with advanced artificial intelligence engines.

The system features multi-tenant architecture enabling multiple hospital organizations to operate isolated environments under a unified platform. It provides 7 role-specific web portals (Super Admin, Hospital Admin, Physician, Nurse, Pharmacist, Lab Technician, Patient) and equips patients with 7 specialized AI engines including multilingual voice triage in regional and global languages (English, Telugu, Hindi, Tamil, Spanish, French, German, Mandarin, Arabic).

2. Problem Statement

Modern healthcare systems face severe operational inefficiencies and technological fragmentation:

1. Data Fragmentation & Siloed Systems

Hospitals frequently use disjointed software for patient registration, pharmacy, and laboratory data, preventing a unified medical timeline across departments.

2. Language Barriers & Clinical Triage Delays

Patients presenting with acute symptoms often face long waiting times for initial symptom triage. Language barriers exacerbate miscommunication, especially in regional healthcare centers.

3. Inpatient Bed & Vital Tracking Friction

Nurses rely on manual logs or outdated spreadsheets to track inpatient bed availability, leading to delayed admissions and ICU bottlenecking.

4. Adverse Drug Interactions & Manual OCR Errors

Manual prescription processing increases the risk of polypharmacy drug-drug contraindications, while handwritten lab reports cause diagnostic interpretation delays.

5. Lack of Security & Infrastructure Monitoring

Hospital IT administrators lack real-time visibility into cyber threat anomalies, HIPAA compliance risks, server node latencies, and revenue leakages.

3. Project Objectives

- Unified Multi-Tenant Platform

Develop a secure Next.js 14 web application supporting isolated multi-hospital tenant scoping (hospitalId) and 7 role-based access control (RBAC) portals.

- Multilingual Speech AI Engine

Implement real-time Speech-to-Text dictation and Dual-Mode Text-to-Speech audio synthesis supporting 9 languages (English, Telugu, Hindi, Tamil, Spanish, French, German, Mandarin, Arabic).

- Real-Time Inpatient Bed Grid & ICU Center

Build an interactive bed management grid (Occupied - Red, Available - Green, Cleaning - Amber) with immediate patient admission/discharge modals and vital sign logging.

- Clinical AI Supercomputer Suite

Integrate 7 patient AI engines (BioBERT Triage, Lab OCR Simplifier, Drug Interaction Checker, Cardiovascular Risk Score, Mental Wellness, Clinical Nutrition) and 6 Super Admin security AI engines.

- Global Floating AI Assistant & Ticket Engine

Deploy a persistent floating robot widget capable of real-time Q&A, voice dictation, and 2-step interactive support ticket creation with screenshot evidence image attachment.

4. System Methodology & Architecture

The system follows a modern full-stack web architecture leveraging Next.js 14 App Router and Prisma ORM with SQLite/PostgreSQL:

1. Multi-Tenant Scoping Layer: Every request undergoes JWT authentication. Database queries dynamically scope data using tenant IDs (hospitalId), ensuring complete tenant data isolation.
2. Speech & Acoustic Processing Layer: Integrates native Web Speech API (webkitSpeechRecognition) for microphone speech dictation. For text-to-speech output, it utilizes browser SpeechSynthesis paired with high-definition Web Audio Synthesis fallback to ensure reliable voice playback for regional languages like Telugu and Hindi.
3. Knowledge & AI Layer: Employs BioBERT vector retrieval-augmented generation (RAG) and dynamic rule-based medical knowledge graphs to generate localized clinical assessments and OTC recommendations.

5. Detailed Explanation of Platform User Roles & Portals

To make the project easy for any new team member or evaluator to understand, the platform is divided into 7 distinct user portals based on real-world hospital workflows:

1. Platform Super Admin (SUPER_ADMIN) [Route: /admin/dashboard]

The master administrator responsible for global platform governance. Super Admins can register new hospital organizations (/admin/hospitals), inspect global user accounts (/admin/users), manage support ticket issues (/admin/support-issues), monitor platform security anomalies, and oversee global operations across all hospital tenants.

2. Hospital Tenant Admin (HOSPITAL_ADMIN) [Route: /admin/dashboard]

The administrator for a specific hospital tenant (e.g. Metropolitan General Hospital). Hospital Admins create clinical departments, onboard physicians and nurses, manage local hospital patient registries, schedule appointments, and monitor pharmacy/lab inventory.

3. Physician / Doctor (DOCTOR) [Route: /doctor/dashboard]

The clinical physician interface. Doctors view their daily patient consultation schedule (/doctor/appointments), open electronic health records (EHR), review patient medical history timelines, author digital prescriptions, and analyze radiology X-ray scans.

4. Nursing Staff (NURSE / STAFF) [Route: /nurse/beds]

The nursing interface focused on inpatient ward care. Nurses manage real-time bed grids (Occupied, Available, Cleaning), log patient vitals (BP, HR, SpO_2, Temperature), record ICU monitoring entries, and execute patient admissions and discharges.

5. Pharmacist (PHARMACIST) [Route: /pharmacy/dashboard]

The pharmacy management interface. Pharmacists receive digital prescription orders, verify medicine stock levels, record dispensing logs, and analyze drug-drug interaction safety risks.

6. Laboratory Technician (LAB_TECH) [Route: /laboratory/dashboard]

The diagnostic laboratory interface. Lab techs process diagnostic test orders, upload/scan paper lab reports using Optical Character Recognition (OCR), and validate lab test results.

7. Patient (PATIENT) [Route: /patient/dashboard]

The patient self-service portal. Patients view their personalized health dashboard, medical timeline, prescription history, lab reports, longevity score, and access 7 specialized AI clinical & voice engines.

6. Specialized Patient Clinical & Multilingual Voice AI Suite

Located at /patient/ai-engines, this suite equips patients with 7 production-grade artificial intelligence engines designed for instant symptom triage, diagnostic interpretation, and voice interaction:

Engine 1: Multilingual Voice AI Recognition & Diagnostic Triage

Allows patients to dictate symptoms via microphone using real-time Speech-to-Text in 9 languages (English, Telugu, Hindi, Tamil, Spanish, French, German, Mandarin, Arabic). The engine analyzes symptoms live and plays back spoken medical responses using Dual-Mode TTS Audio Synthesis.

Engine 2: BioBERT RAG Symptom Triage & OTC Guide Engine

Utilizes transformer vector retrieval-augmented generation (RAG) backed by PubMed citations to perform clinical symptom differential diagnosis and provide safe over-the-counter (OTC) care guidance.

Engine 3: Lab OCR Report Simplifier

Parses dense diagnostic test parameters (HbA1c, Fasting Glucose, Lipid Profile, WBC, Hemoglobin) and converts medical jargon into plain-text patient explanations and risk flags.

Engine 4: Polypharmacy Drug Interaction & Safety Checker

Analyzes multi-drug combinations (e.g. Warfarin + Ibuprofen + Paracetamol) to detect potential drug-drug interactions, toxicity risks, and contraindications.

Engine 5: Chronic Cardiovascular Risk Score Engine

Calculates 10-year Framingham and ASCVD cardiovascular risk percentages dynamically based on age, BMI, systolic/diastolic BP, blood glucose, and exercise frequency.

Engine 6: Mental Wellness & Sleep Protocol Generator

Evaluates stress levels and sleep duration to generate personalized cognitive wellness recommendations and sleep hygiene protocols.

Engine 7: Clinical Nutrition & Meal Plan Generator

Generates disease-specific 7-day meal protocols for conditions such as Type 2 Diabetes, High Blood Pressure, Hyperlipidemia, and Kidney Health.

7. Super Admin Infrastructure & Security AI Suite

Located at /admin/ai-engines, this suite provides Super Admins with 6 operational governance engines:

1. Cyber Threat Anomaly Detector
Monitors platform IP traffic, DDoS attack vectors, and brute-force login attempts in real time.

2. Hospital Compliance & HIPAA Auditor
Audits multi-tenant data privacy risks, patient consent logs, and regulatory compliance ratings.

3. Healthcare Revenue Predictor
Forecasts hospital monthly revenue based on inpatient bed occupancy rates and insurance claim approvals.

4. System Server Health Monitor
Inspects server node latency, database connection pool saturation, and Redis cache hit ratios.

5. AI Model Governance Auditor
Tracks clinical AI model drift, calculates hallucination indices, and logs prediction safety audits.

6. Support Ticket Auto-Triage NLP
Classifies incoming hospital support tickets by sentiment, urgency, and priority level.

8. Global Floating Nexo AI Assistant & Support Ticket Engine

Rendered persistently across the entire platform via FloatingAiChatbot.tsx:
- Microphone Voice Dictation: Patients/staff can speak queries from any page.
- Voice Audio Toggle: Enables or mutes spoken audio response playback.
- Interactive Support Ticket Creation: Allows users to submit official support tickets (TICKET-2026-XXXX) capturing Contact Name, Mobile Phone, Email Address, Category, Subject, Description, and Screenshot Evidence Image attachment.
- Live Synchronization: Automatically syncs tickets to /admin/support-issues for admin processing.

9. Implementation Details & Database Schema

The project is implemented using Next.js 14 TypeScript App Router. Key core files include:
- src/components/NavigationSidebar.tsx: Dynamic sidebar rendering role-specific sitemaps.
- src/components/FloatingAiChatbot.tsx: Global floating assistant with voice dictation & ticket creation.
- src/app/patient/ai-engines/page.tsx: Patient 7 AI engines suite with speech recognition & audio synthesis.
- src/app/admin/ai-engines/page.tsx: Platform Super Admin security AI suite.
- src/app/nurse/beds/page.tsx: High-contrast inpatient ward bed grid.
- src/app/api/admin/ai/route.ts: Central real-time AI backend API route supporting dynamic queries & target language translations.

Database Schema Summary (Prisma 55+ Relational Models)

The SQLite/PostgreSQL database (dev.db) structures data across 55+ relational entities:
- Tenant & User Models: Hospital, Department, User, Role, Permission, RolePermission, UserPermission
- Clinical Staff Profiles: StaffProfile, DoctorProfile, NurseProfile, Patient
- Medical Records & Encounters: MedicalRecord, Appointment, Consultation, Prescription, Medicine, PharmacyInventory, DispensingLog
- Diagnostics: LabTest, LabOrder, LabReport, MedicalImage
- Inpatient & ICU: Ward, Bed, Admission, Discharge, ICUUnit, ICUBedLog
- System & Security: AIPredictionLog, SupportIssue, AuditLog

10. Experimental Results & Key Accomplishments

1. Clean Compilation: Verified via "npx tsc --noEmit" with 0 TypeScript compilation errors.
2. 100% Real-Time AI Responses: Replaced mock timers with live query processing at /api/admin/ai.
3. Universal Multilingual Audio Fallback: Solved regional voice playback silent errors on Windows by implementing Web Audio Synthesis fallbacks for Telugu, Hindi, Tamil, and Spanish.
4. High-Contrast Accessible Design: Upgraded typography to font-black and slate-900 high contrast for nurse bed cards, prescriptions, and medical tables.

11. Future Scope & Enhancements

1. IoT Wearable Integration: Connect pulse oximeters and continuous ECG monitors for real-time telemetry.
2. HL7 / FHIR Standardization: Implement international healthcare data interchange protocols.
3. Federated AI Learning: Enable privacy-preserving clinical AI model training across isolated hospital tenants.

12. Project Execution & Pre-configured Demo Accounts

To run locally: cd Project -> npm install -> npx prisma db push -> npm run db:seed -> npm run dev -> http://localhost:3000
Database GUI: npx prisma studio -> http://localhost:5555

Default Password for all accounts: password123

Role	Email	Route	Scope
Super Admin	superadmin@nexomedico.ai	/admin/dashboard	All Platform Tenants
Hospital Admin	admin@metrohospital.com	/admin/dashboard	Metropolitan Hospital
Doctor	dr.smith@metrohospital.com	/doctor/dashboard	Cardiology / OP
Nurse	nurse.sarah@metrohospital.com	/nurse/beds	Inpatient Wards
Pharmacist	pharma.alex@metrohospital.com	/pharmacy/dashboard	Hospital Pharmacy
Lab Tech	lab.tech@metrohospital.com	/laboratory/dashboard	Clinical Lab & OCR
Patient	john.doe@gmail.com	/patient/dashboard	Personal Health Portal